

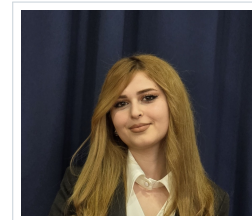
Ghinwa Ismail

PhD Researcher in 5G Network Digital Twins

Strasbourg, France | gismail@unistra.fr

Portfolio | GitHub | LinkedIn | Google Scholar

ORCID: 0009-0002-3306-007X



Academic Profile

PhD researcher at the University of Strasbourg's ICube Laboratory, working on trustworthy Network Digital Twins for 5G systems. Research combines trace-driven traffic modelling, reproducible OpenAirInterface experimentation, radio-condition replay, and uncertainty-aware network performance evaluation. Background spans telecommunications engineering, data science, distributed systems, and experimental wireless networking.

Research Interests

5G and beyond | Network Digital Twins | O-RAN and intelligent RAN control | Traffic modelling | Wireless experimentation | Machine learning for networks | Reproducibility and uncertainty quantification

Research Experience

PhD Researcher in Telecommunications

Nov. 2024–Present

University of Strasbourg, ICube Laboratory, Strasbourg, France

- Develop trustworthy 5G Network Digital Twin workflows linking realistic traffic generation, controlled experimentation, and KPI estimation.
- Designed a lightweight trace-driven burst model using flow-aware segmentation, HDBSCAN burst types, Markov transitions, and distributional validation.
- Conduct experiments with OpenAirInterface 5G Standalone, RFSim, O-RAN/nRT-RIC components, Colosseum, POWDER and Docker.
- Investigate measurement-driven replay of radio conditions and uncertainty-aware prediction of throughput, latency, resource usage, and packet loss.

Research Intern

Feb.–Aug. 2024

Orange Innovation, Caen, France

Designed an API for Ethereum smart contracts and validated it through a distributed-communication prototype. Managed GitLab CI/CD and Kubernetes-based delivery workflows.

Publications

- G. Ismail**, S. Si-Mohammed, and F. Théoleyre, “Lightweight Trace-Driven Burst Traffic Generation for 5G Network Digital Twins,” in *Proc. IEEE NetSoft*, Berlin, Germany, 2026, pp. 162–170. doi:10.1109/NETSOFT70012.2026.11603454.
- G. Ismail**, S. Si-Mohammed, and F. Théoleyre, “Demo: TwinDash – a Dashboard-Based Traffic Generator for Reproducible 5G Experiments,” accepted at the *51st IEEE Conference on Local Computer Networks (LCN 2026)*, Coimbra, Portugal, forthcoming.
- G. Ismail**, J. Hatin, J. Cantais, and V. André, “A Decentralised Videoconferencing Service using Ethereum Smart Contracts,” in *Proc. BRAINS*, 2024. doi:10.1109/BRAINS63024.2024.10732428.

Education

PhD in Telecommunications

Nov. 2024–Present

University of Strasbourg, ICube Laboratory, France

Doctoral supervisors: Prof. Fabrice Théoleyre and Dr. Samir Si-Mohammed.

MSc in Data Science and Network Intelligence

2023–2024

Télécom SudParis, Institut Polytechnique de Paris, France

Double-degree programme. Final grade: 17/20.

MSc in Telecommunication Engineering

2022–2024

University of Calabria, Italy

Final grade: 108/110.

BSc in Telecommunication Technologies

2016–2020

Tishreen University, Syria

Final grade: 90.5%; ranked first in the cohort in each of the four academic years.

Teaching and Supervision

Main Supervisor, Master’s Internship

2026

University of Strasbourg, France

Supervised Jeronimo HERDOIZA FUSTILLOS on “Deployment and Benchmarking of a Reproducible 5G Standalone Platform.” Guided the deployment, automation, controlled experimentation, and KPI evaluation of a reproducible 5G workflow.

Teaching Assistant

2020–2022

Tishreen University, Syria

Delivered tutorials and practical sessions in Electronic Measurements, Network Management, Security, and computer skills, using OptiSystem and VMware simulations.

Selected Research and Engineering Contributions

Trace-driven traffic generation and TwinDash

2025–2026

Developed application-specific burst traffic models and a dashboard-oriented workflow for configurable, reproducible 5G experiments; released code and experimental artifacts publicly.

Measurement-driven 5G radio-condition replay

2025–Present

Designed a validation-led workflow for replaying measured radio variations in OAI RFSim, including measurement-compatibility checks, calibration, repeatability analysis, and continuous probing.

Decentralised videoconferencing service

2024

Built and evaluated an Ethereum smart-contract API supporting secure distributed communication services during the Orange Innovation internship.

Technical Skills

Programming	Python, C++, MATLAB
5G & networking	OpenAirInterface 5G SA, O-RAN, near-RT RIC, SDN/NFV, TCP/IP, SIP, MQTT, VoIP
Experimentation	RFSim, Colosseum, POWDER, MGEN, iperf3, tcpdump, KPI collection and analysis
Infrastructure	Linux, Git/GitLab, Docker, Kubernetes, CI/CD, VMware
ML methods	HDBSCAN, Gaussian mixture models, Markov models, Random Forest, XGBoost, SVC

Awards and Distinctions

- **Best Poster Award**, Journée ICube, for “Toward Trustworthy Digital Twins for 5G Networks,” 2 July 2026.
- **Ranked first in the cohort in every year of the four-year BSc programme**, Tishreen University, 2017–2020.
- **AL-Basil Outstanding Graduate Certificate**, Tishreen University, 2020.
- Honoured participant, Al-Basel Fair for Innovation, 2021.

Selected Academic Activities

- Presented the full paper “Lightweight Trace-Driven Burst Traffic Generation for 5G Network Digital Twins” at IEEE NetSoft 2026, Berlin, Germany.
- SLICES-RI / CONVERGE Summer School, INESC TEC, Porto, Portugal, 25–27 June 2025: hands-on training in 6G, Open RAN, Network Digital Twins, OpenAirInterface, and OAIBOX.

Languages

Arabic (native) | English (advanced) | French (beginner)